DEIVA MANI KAILASAM

Senior Full Stack Software Engineer – Cloud-native, Distributed Systems & GenAI

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PROFILE

Senior Full Stack Software Engineer with 8+ years building cloud-native, multi-tenant platforms and data-intensive web applications on AWS/Azure using Java, Spring Boot, C#, Python, TypeScript, and modern frontend frameworks. Led design and delivery of distributed microservices handling 50K+ daily API requests and platforms serving 500K+ users with strong observability, reliability, and CI/CD automation. Strong experience in GenAI (RAG, embeddings, LLM-powered analytics) and integration of AI into production features and developer workflows. Proven technical leadership: architecture ownership, cross-team collaboration, and mentoring engineers while driving coding standards and best practices.

PROFESSIONAL EXPERIENCE

SENIOR SOFTWARE ENGINEER — Aug 2021 - Present

Kinesso – Omnicom (Earlier IPG)

Platform & Backend

- Architected and led a multi-module Spring Boot 3.x microservices platform with JWT/OAuth2, multi-tenant data access, and OpenAPI contracts, serving 50K+ daily API requests across hybrid SQL/NoSQL stores (PostgreSQL, SQL, Snowflake, MongoDB).
- Designed multi-tenant patterns (DB-per-tenant in PostgreSQL/SQL Server, collection-per-tenant in MongoDB, row-level security) ensuring zero cross-tenant leakage across 12+ environments.
- Implemented a gRPC-based plugin architecture for integrating Python microservices and client-specific React dashboard plugins, improving onboarding speed for new services.
- Reduced 99th percentile latency by 55% through large-payload handling with Spring WebClient, Python asyncio, MongoDB GridFS, and parallel query execution plus hybrid caching (Caffeine, Redis, MongoDB).

Frontend & UX

- Built and maintained a multi-framework design system with 40+ reusable components (Angular, React, Next.js) distributed via private npm (Nexus), enabling consistent UX across 6 internal products.
- Implemented advanced AG Grid Enterprise tables (filtering, sorting, row expansion, CSV/Excel export via ExcelJS) and GenAI-powered insights, enabling non-technical users to self-serve analytics.
- Delivered a white-label theming system (Angular SCSS + React/Next.js Tailwind) supporting 5+ brand variants with dynamic stylesheet injection, significantly reducing theme configuration effort.
- Authored Storybook documentation (Angular/React/Next.js, MDX, Compodoc) with 150+ stories, a11y tests, and GenAI component suggestions, increasing component discoverability by 85%.

Data, AI & Analytics

- Built a hybrid data access layer spanning PostgreSQL, SQL Server, Snowflake, MongoDB with Liquibase migrations and MongoDB schema versioning, simplifying cross-database querying.
- Integrated Google Ads, Meta/Facebook, Bing Ads, Adverity, Snowflake, Power BI, and MongoDB Atlas Search with OAuth2/OIDC and Python data connectors, enabling near real-time advertising analytics.
- Designed GenAI pipelines (RAG, embeddings per tenant, LLM-based anomaly detection) and a Python-based data quality scoring engine, reducing undetected data quality issues by 75%.

Frontend & UX

- Established automated testing across 1000+ unit tests (Jest, Vitest, Pytest), 100+ integration/E2E tests (Cypress, Playwright), and controller-level tests (MockMvc, Pytest), achieving 80%+ coverage across Java/Python/UI.
- Configured CI/CD with Jenkins and GitHub Actions for React/Next.js using multi-stage Docker builds, SonarQube gates, and PR validation, cutting merge-to-deploy time by 40%.
- Designed Kubernetes infrastructure (Spring Boot, Python FastAPI, Next.js) with ConfigMaps, Vault secrets, MongoDB StatefulSets, liveness/readiness/startup probes, and Datadog/MongoDB dashboards, reducing MTTR by 65%.
- Developed Terraform IaC for AWS ECS/Fargate and MongoDB Atlas with GitOps via ArgoCD, enabling one-command deployments and eliminating manual infra changes.
- Mentored 5+ engineers (3 promoted to mid-level) and drove multi-language coding standards.

SOFTWARE DEVELOPER
Dec 2020 - Jul 2021

ACE Contractors Pvt Ltd

- Architected and owned a Progressive Web App (PWA) platform for construction operations on Azure (App Service, Azure PostgreSQL Flexible Server, Functions, Blob Storage, AD B2C), achieving 99.9% uptime for 200+ daily active users across 10+ sites.
- Built mobile-first Angular frontend with PWA (Service Workers, App Manifest, Cache API), enabling offline workflows for field workers and improving daily task completion rates by 40%.
- Implemented Node.js/Express backend with gRPC microservices, stream processing for real-time timesheet validation and contract updates, and sub-100ms response times for 95th percentile requests.
- Designed PostgreSQL schemas for employees, projects, contracts, timesheets, and safety incidents using JSONB and full-text search for fast retrieval.
- Delivered CI/CD via Azure DevOps (YAML), multi-stage Docker builds, Azure Container Registry, ARM/Bicep templates, cutting deployment time from 2 hours to 12 minutes (94% reduction).
- Led weekly exec syncs with the company owner, prioritising high-ROI features that reduced payroll disputes, digitised timesheets, and accelerated compliance reporting 3x for WorkSafe Victoria.

DEVELOPER
Jan 2016 - May 2019

Tata Consultancy Services

- Led migration of a legacy Java EE monolith to Spring MVC + Spring Boot layered architecture (Repository–Service–REST Controller), reducing technical debt by 70% and removing 15+ critical security vulnerabilities.
- Gradually migrated AngularJS 1.5 to Angular 5, implementing RxJS, Angular Router, Angular Material, and lazy loading, achieving 3x faster initial page loads and 65% bundle size reduction.
- Designed PostgreSQL schemas (replacing Oracle) for projects, employees, timesheets, and billing with stored procedures and indexing, cutting query times by 65% and improving page load times.
- Implemented test-first development with JUnit/Mockito/TestNG and Selenium WebDriver E2E tests, achieving 85%-unit test coverage and catching 25+ regressions early.
- Containerized Spring Boot services using Docker multi-stage builds, simplifying deployments and enabling 8+ major releases that consolidated multiple legacy systems.

PROJECTS

- Feature-Rich UI Component Library – Multi-framework design system (Angular, React, Next.js) with 40+ reusable components, used across multiple internal products.
- AI Agent for Code Generation – LLM-powered tool (GPT-4, LangChain) that generates boilerplate and integration code, accelerating development for internal teams.
- Edge Water Management Platform – Full-stack web app for smart water monitoring and optimisation with modern Angular + cloud backend.

SKILLS

- **Languages & Frameworks**
  Java, Python, TypeScript, JavaScript, C#; Spring Boot, Spring, Node.js/Express, Angular (Latest), React, Next.js, Tailwind, RxJS.
- **Cloud, Data & DevOps**
  Azure (App Service, Functions, PostgreSQL Flexible Server, Azure DevOps), AWS (ECS, Fargate, EC2, Lambda), Docker, Kubernetes, Terraform, ARM/Bicep, Jenkins, GitHub Actions, Azure DevOps, Git.
- **Data & Messaging**
  PostgreSQL, SQL Server, Snowflake, MongoDB, MongoDB Atlas Search, Kafka; Redis, Caffeine; OpenAPI/Swagger, gRPC, REST.
- **AI & Practices**
  GenAI (LLMs, RAG, embeddings, prompt engineering, LangChain), test automation (JUnit, Jest, Cypress, Pytest), observability (Datadog, Application Insights), engineering practices (clean code, code review, performance tuning, incident response).
- **Architecture & Design**
  System Design, Microservices and distributed systems, multi-tenant SaaS, API design, event-driven architectures (CQRS, saga), reactive programming, SQL/NoSQL data modelling.

EDUCATION

Master of Technology in Information Technology
Aug 2019 - May 2021
La Trobe University, Australia

Bachelor of Technology in Information Technology
Aug 2011 - May 2015
Valliammai Engineering College, India

CERTIFICATIONS

- Google Cloud Certification on GenAI (2025)
- IBM Certification on Prompt Engineering (2025)

AWARDS

- Best IT Project Award - La Trobe for the best University project (2020)
- Pursuit of Knowledge Award – TCS For knowledge sharing (2018)
- On the Spot Award – TCS For exceptional project delivery (2018)
- Cornerstone Award – TCS For best performance in the project (2017)
- Star Award – TCS For the best employee during training period (2016)

LANGUAGES

- English | Fluent
- Tamil | Native
- Hindi | Moderate